

Keerti | teaches | at | Educosys

↓
↳ Tokens → ① Tokenization

② Encoding → label
→ one hot Encoding
OHE

Keerti	[1, 0, 0, 0]
teaches	[0, 1, 0, 0]
at	[0, 0, 1, 0]
Educosys	[0, 0, 0, 1]

} → No meaning

③ Embedding → word2vec, GloVe

Parallel $\Rightarrow$ Embedding + Positional Encoding

(P1)
keeki

(P2)
Teacher

(P3)
at

(P4)
Education

[0.2 0.3 0.4 0.8 0.1 0.2]

Dimension of embedding = 6

$$PE(pos, 2i) = \sin\left(\frac{pos \cdot 2i / d_{model}}{10000}\right)$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos \cdot 2i / d_{model}}{10000}\right)$$

pos = 0 $\Rightarrow$ keeki
i = 0

Embedding $\Rightarrow$ 6
PE's dimension

[0, 1, 2, 3, 4, 5]

PE

Keerti

pos = 0

Teachu

pos = 1

at

pos = 2

Educoys

pos = 3

i = 0

1

2

3

4

5

$$\sin\left(\frac{0}{1000} = 0/d\right)$$

$$\cos\left(\frac{0}{1000} = 0/d\right)$$

$$\sin\left(\frac{0}{1000} = 0/d\right)$$

$$\cos\left(\frac{0}{1000} = 0/d\right)$$

$$\sin\left(\frac{0}{1000} = 0/d\right)$$

i = 0

i = 1

i = 2

i = 3

i = 4

i = 5

0.2

0.3

0.1

0.06

0.7

0.2

$$\sin\left(\right)$$

$$\cos\left(\right)$$

0.1

4 words

Embedding $\rightarrow$ 6 dimension

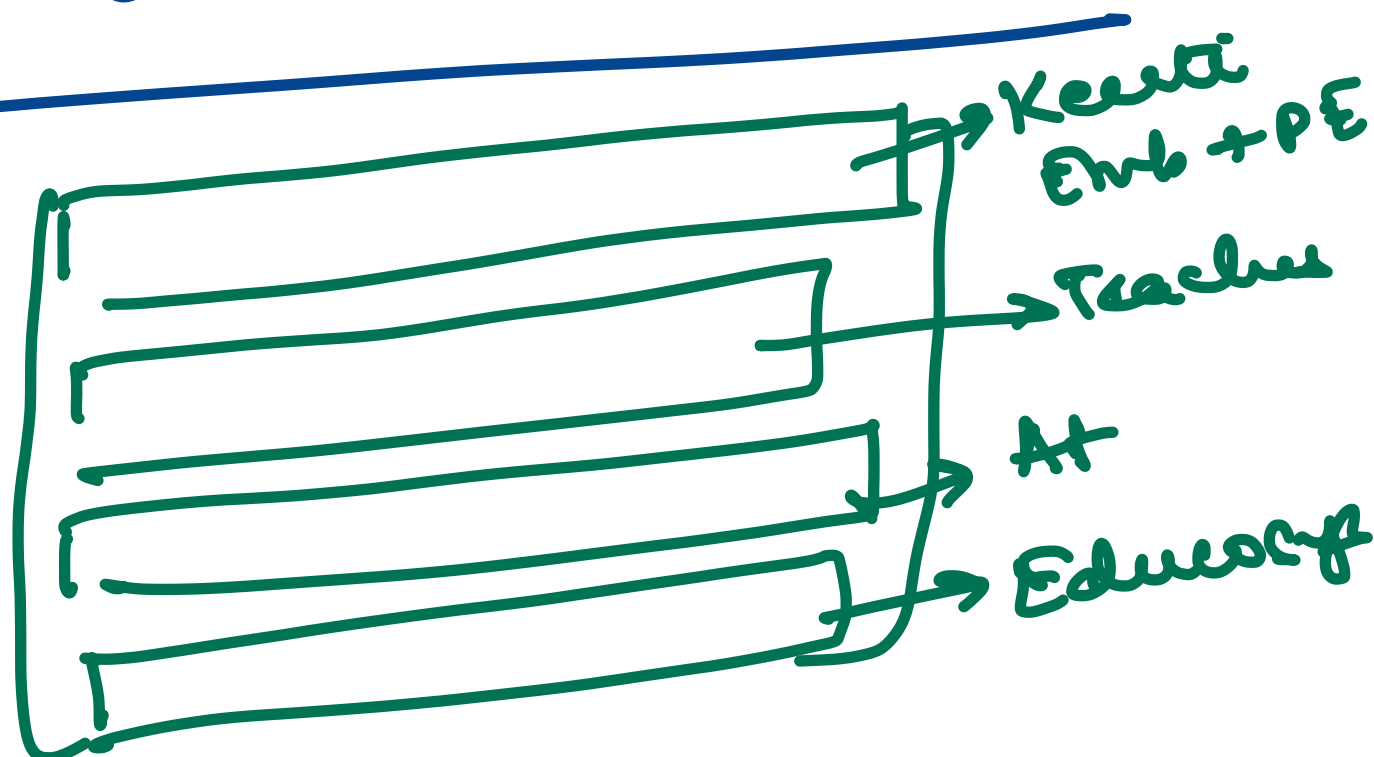
+ PE $\rightarrow$ 6 dim

$\rightarrow$ 6 dimension

Input $\rightarrow 4 \times 6$

Rows $\rightarrow 4 \rightarrow$ 4 words

Columns $= 6$
 $\rightarrow$ Dimension



4 * 6

Input vector

x Wt Matrix $\Rightarrow Q, K, V$

keys

0.2	0.3	0.1	0.5	0.02	0.7
0.1	0.2	0.6	0.4	0.3	0.8
0.9	0.4	0.3	0.2	0.1	0.2
0.6	0.5	0.4	0.3	0.2	0.1

4 * 6

Wt matrix

w_1	w_2	w_3	w_4	w_5	w_6
w_1	-	-	-	-	-
w_2	-	-	-	-	-
w_3	-	-	-	-	-
w_4	-	-	-	-	-
w_5	-	-	-	-	-
w_6	-	-	-	-	-

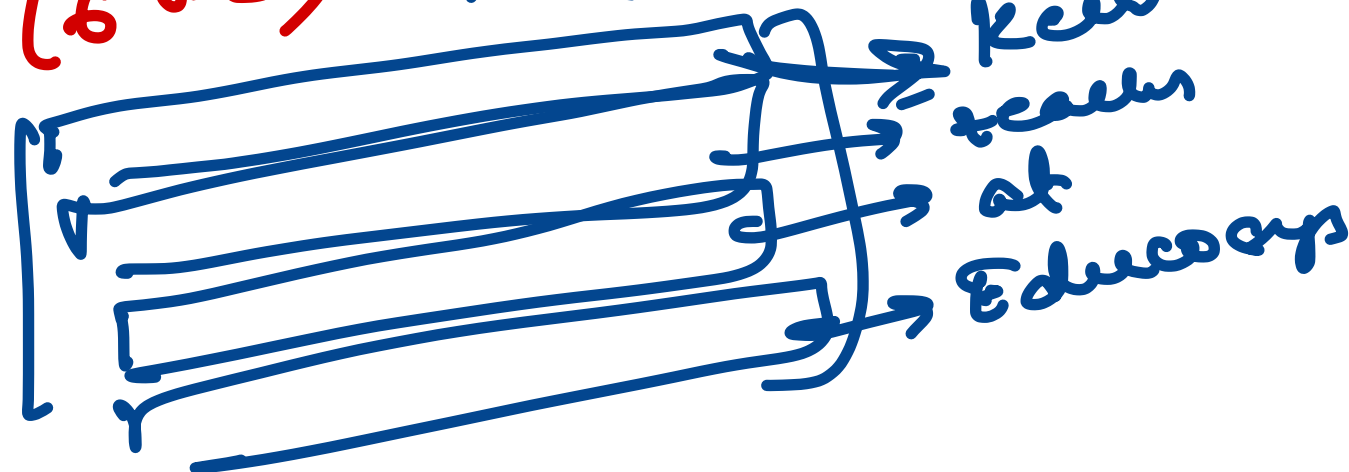
Wt matrix

Rows = 6

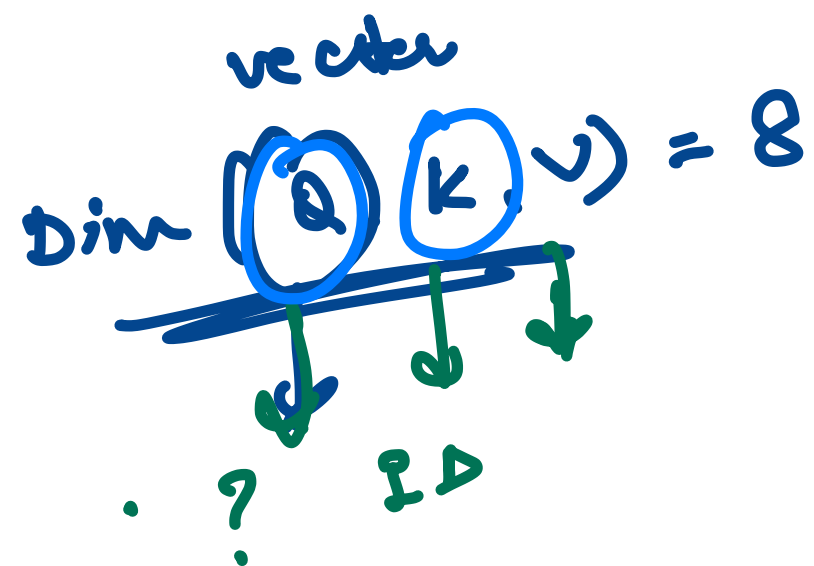
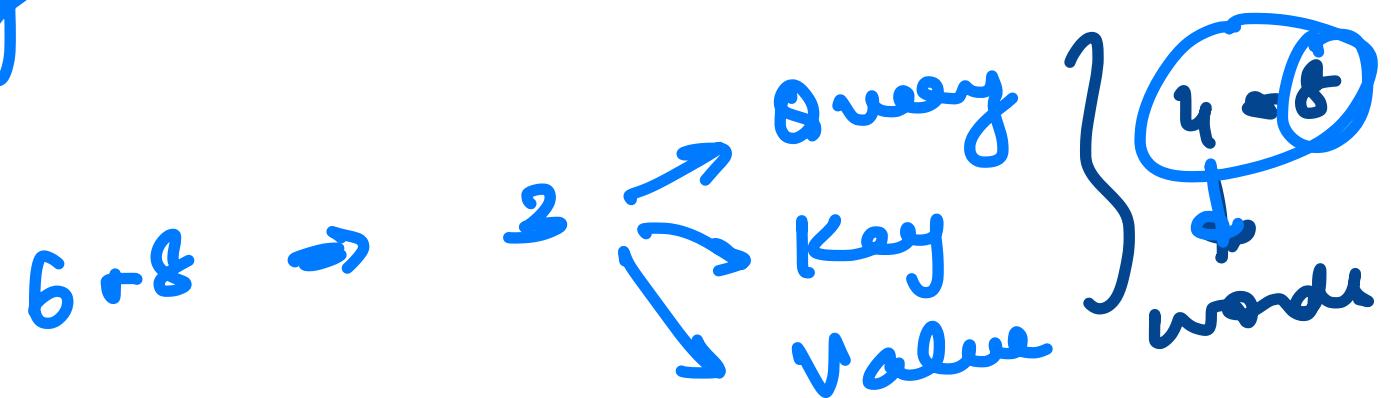
Hyperparameter
cols = 8

$$(4 * 6) (6 * 8) \Rightarrow (4 * 8)$$

Query
Matrix

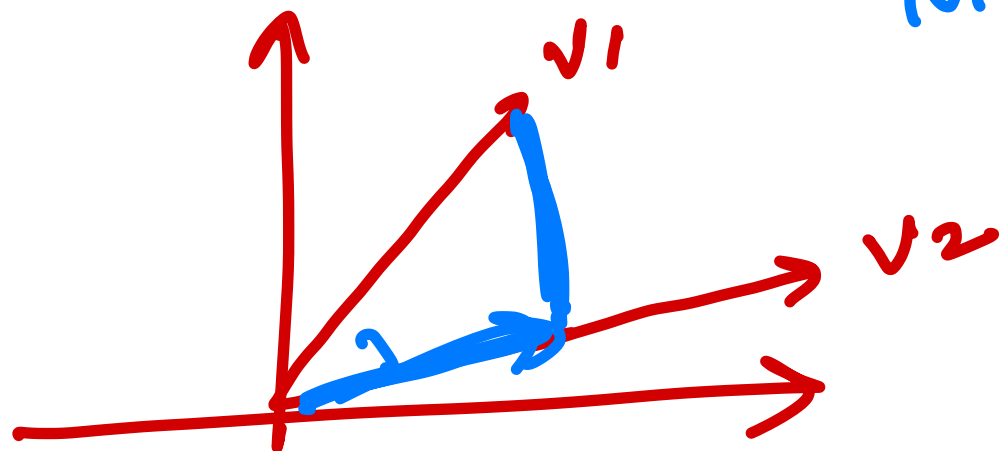


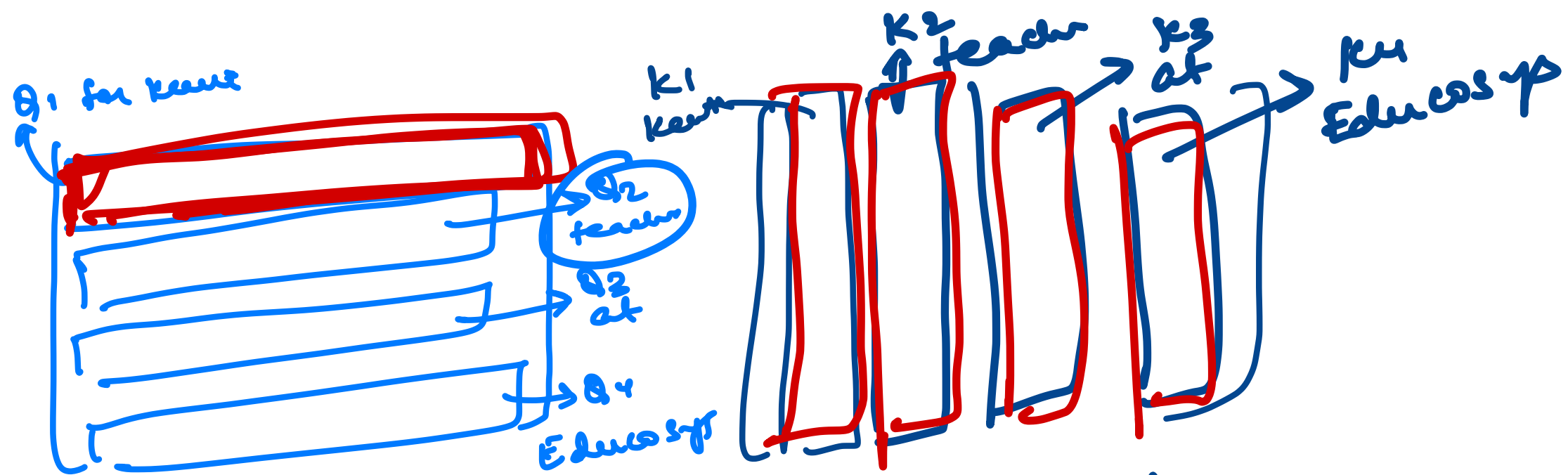
Key Matrix, Value Matrix



(Q, K) $\Rightarrow$ Dot Product

$$|v_1| |v_2| \cos \theta$$





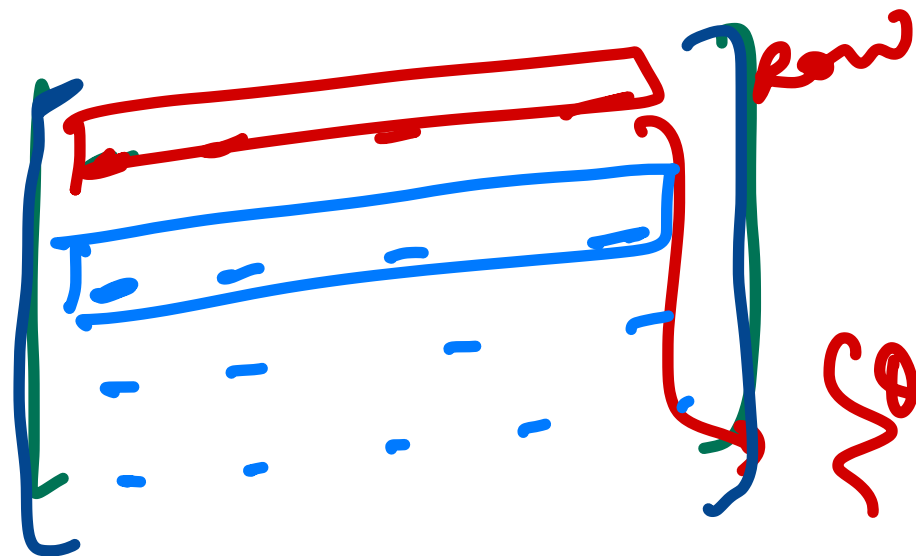
$$(4 \times 8)$$

$$(8 \times 4) \text{ (key) }^{\text{Transpose}}$$

$$(4 \times 8) (8 \times 4) \Rightarrow (4 \times 4)$$

$$Q, K^T$$

no. of keys



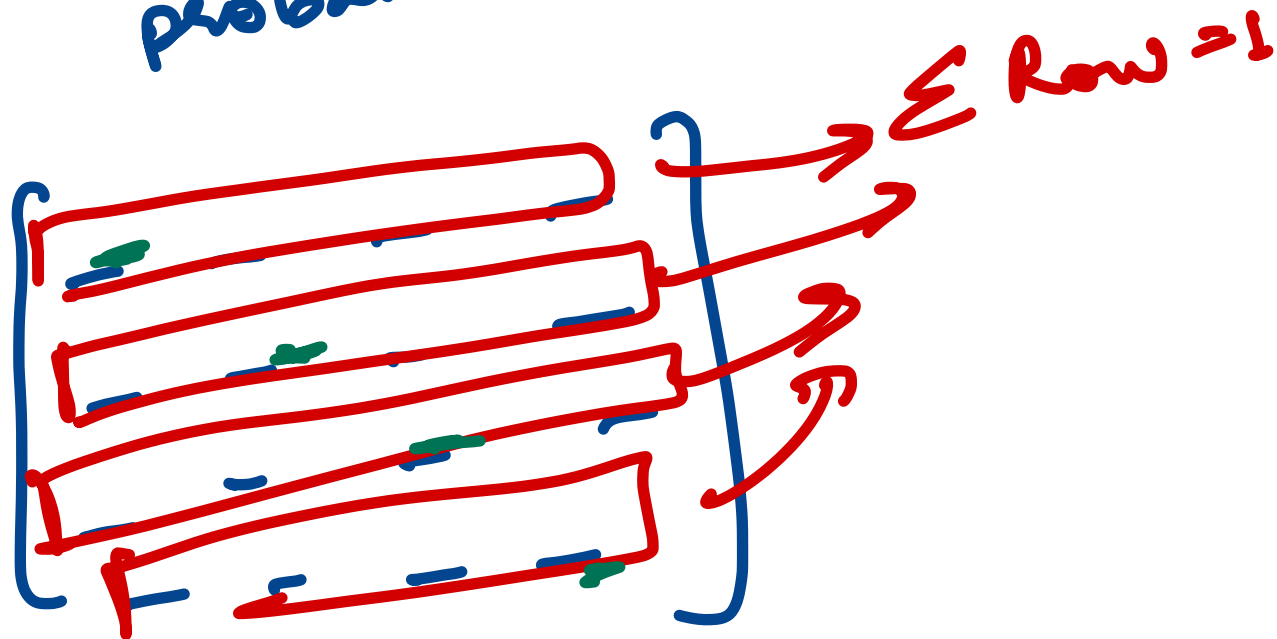
$Q_1 \rightarrow \text{Keen}$
 $K \rightarrow \text{Keen}$
 Query taken attending to
 keys of all tokens

$$\frac{Q_k^T}{\sqrt{d_k}} \rightarrow \text{dimension of } Q_k/v \rightarrow \text{Scaling}$$

$$\Downarrow$$

$$\text{softmax}\left(\frac{Q_k^T}{\sqrt{d_k}}\right)$$

probab such that total prob = 1

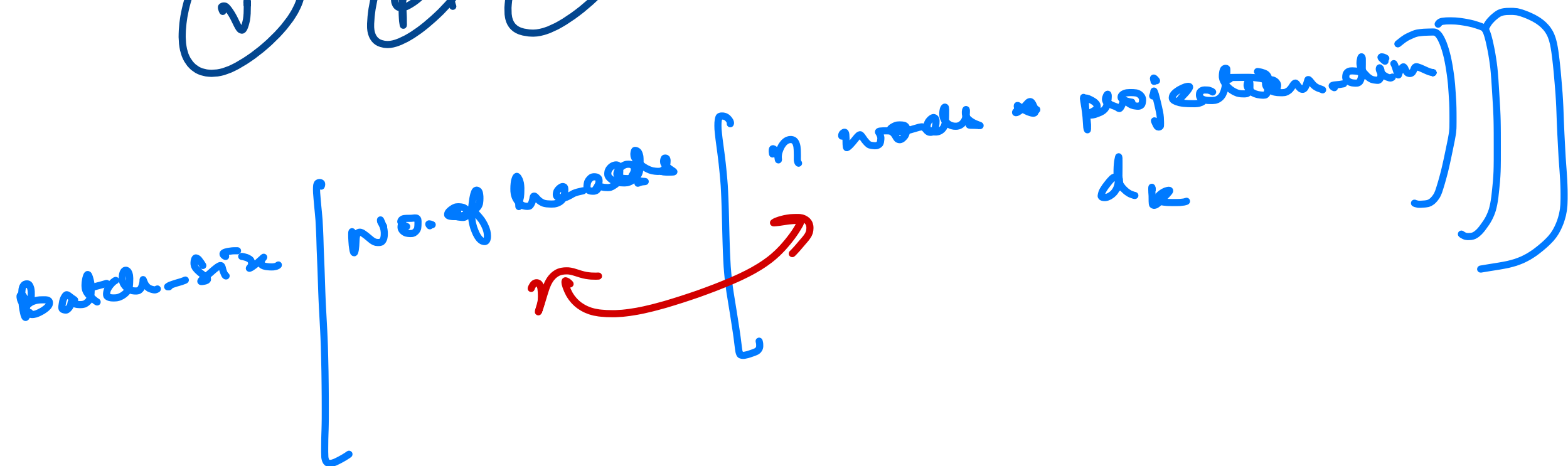
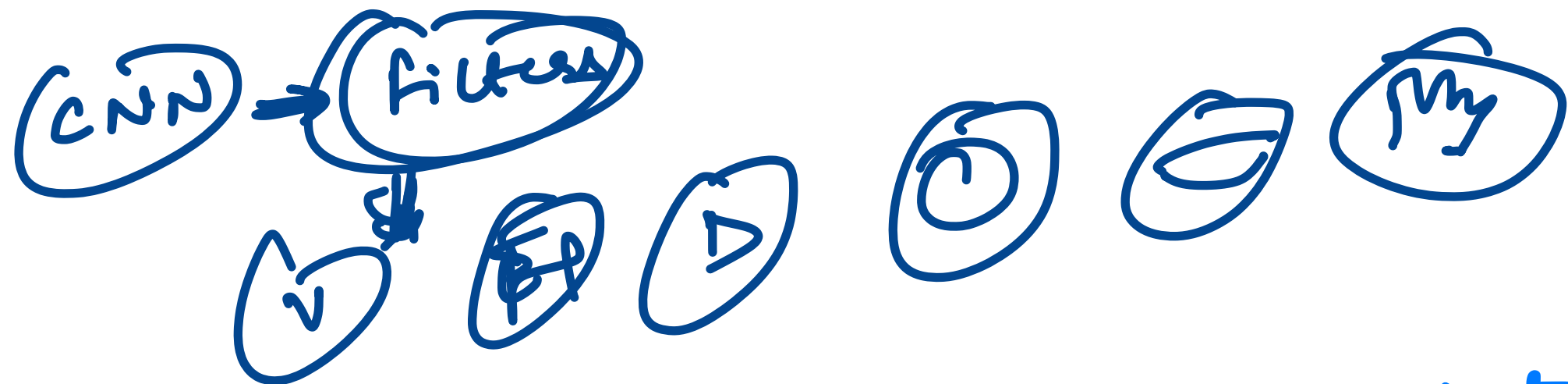


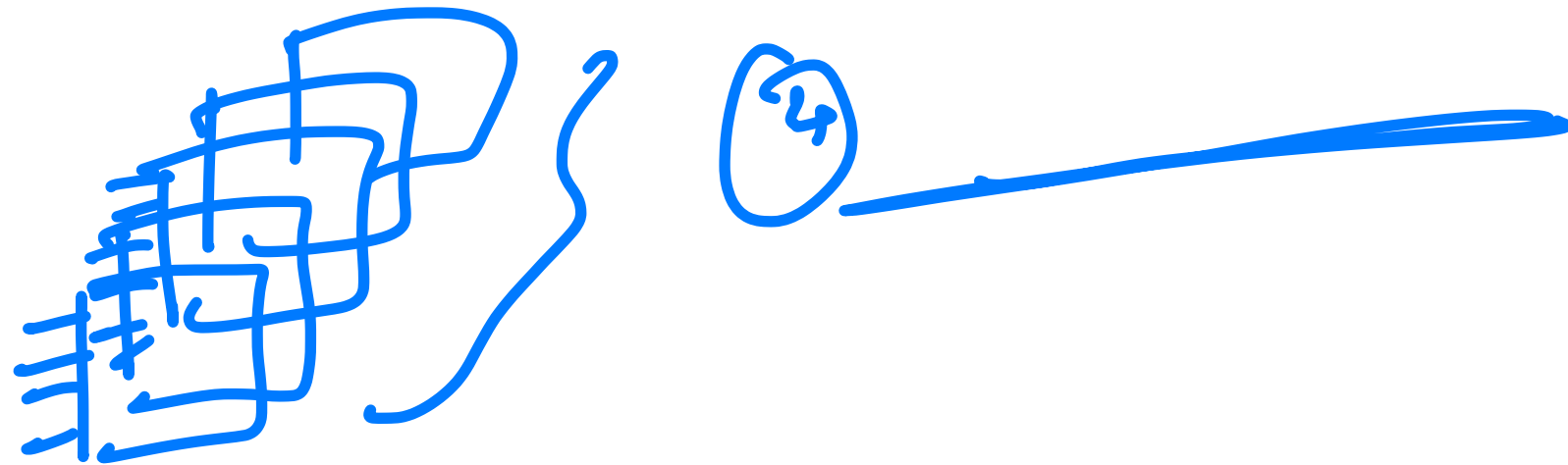
$$\text{Self Attention Score} = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) \cdot V_{\text{netin}}$$

↓
4+8

Self Attention Score

cat sat on the mat
 I
 noun
 cat → animal





Decoder → Autoregressive
Self depends on past values

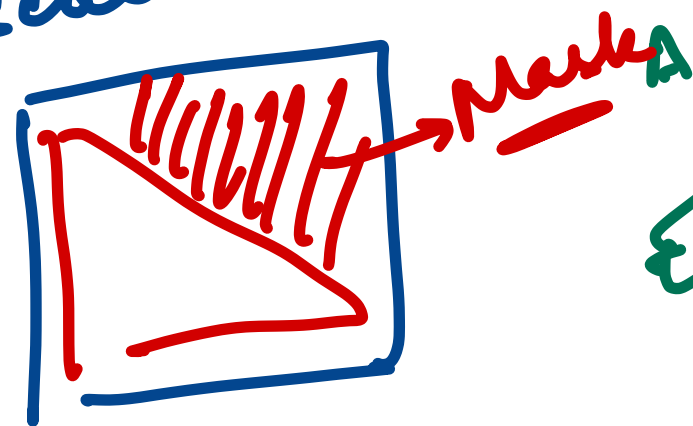
- - - - ↑ ↑ ↑ ↑

RNN, LSTM, Decoder → Autoregressive

Diff b/w Encoder, Decoder

$$\frac{QK^T}{\sqrt{d_k}}$$

scaled score

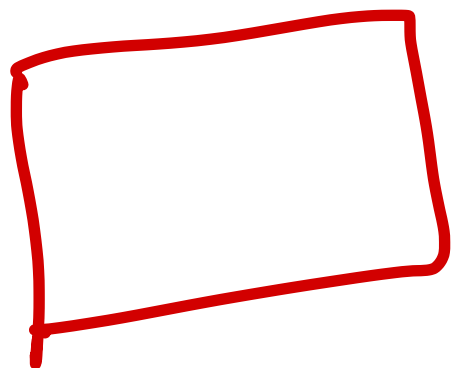


	K	T	A	E
K	⊗	⊗	⊗	⊗
T	⊙	⊙	⊗	⊗
A	⊙	⊙	⊙	⊗
E	⊙	⊙	⊙	⊗

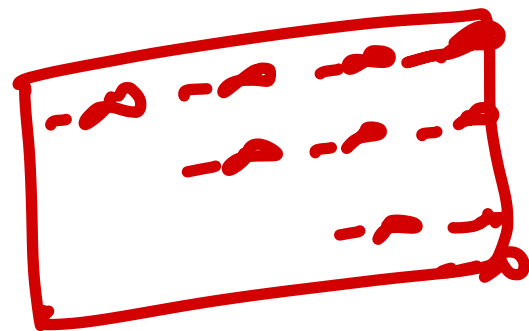
Keerti teaches at Edurexp.

0 Attention Score for future tokens

$$\frac{QK^T}{\sqrt{d}}$$



+



= Masked Scores

Softmax($-\infty$)

= 0

(e^{-2})
, 1, ...

Softmax(Masked Scores)

Decoder without masking